AI-Powered Collections System: Workflow, Governance & Rollout

> **Project:** AI-Driven Delinquency Risk Assessment for Geldium's Credit Card Collections
> **Document 4 of 4:** End-to-End System Design (from the Collections strategy deck)

Overview

The first three documents covered the data, the model, and the business recommendation. This document describes how everything comes together as an **operational, continuously learning collections system** — what runs automatically, where humans stay in control, how fairness is ensured, the expected benefits, and how it will be rolled out.

1. End-to-End AI-Powered Collections Workflow

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1. Data Ingestion → 2. Risk Prediction Model → 3. Decision Engine → 4. Intervention & Feedback Loop
        ↑                                                                          │
        └──────────────────────── model retrains on feedback ──────────────────────┘
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Stage	What Happens
1. Data Ingestion	Customer data — repayment history, credit utilisation, income, etc. — with **real-time updates from transaction systems**
2. Risk Prediction Model	Assesses delinquency likelihood using the key predictors and **outputs a risk score for each customer**
3. Decision Engine	Drives **outreach prioritisation** and **intervention logic**
4. Intervention & Feedback Loop	Actions are initiated and monitored; the **model retrains using performance feedback**

2. Continuous Learning Loop for Smart Collections

The system improves itself over time:

- **Track outcomes** of interventions (e.g., response rates, repayments).
- **Adjust decision thresholds** and outreach strategies based on results.
- **Flag underperforming segments** for review.
- **Retrain the model every 30–60 days** using the feedback loop.

3. Autonomous vs. Human-in-the-Loop Decisioning

Autonomous (AI handles)	Human Oversight (people handle)
Predict delinquency risk using AI models	Review edge cases
Prioritise outreach (e.g., SMS, email)	Approve high-impact interventions
Trigger standard interventions	Monitor fairness and audit outcomes
Update the model using feedback data	Escalate disputes or customer appeals

Principle: Routine, high-volume, low-risk actions are automated for speed and scale; sensitive, high-impact, or disputed decisions stay with humans.

4. Ensuring Fairness, Transparency, and Compliance

Pillar	How It's Achieved
Fairness Monitoring	Assess bias across demographic groups
Explainability	Use interpretable models and SHAP values
Human Review	Manual approval for sensitive actions
Compliance	Align with financial regulations and privacy laws

5. Anticipated Benefits

5.1 Business KPIs

KPI	Expected Impact
30-day+ delinquency rate	**15–25% reduction**
Outreach efficiency	**30% improvement**
Operating cost	Reduced through automation
Repayment rates	Higher, through targeted nudges

5.2 Customer Outcomes

- More **personalised communication**
- **Support for financially stressed** users
- **Transparent, explainable** decisions
- Improved **satisfaction and retention**

6. Phased Implementation Strategy

Phase	Focus
Phase 1	Data audit and feature engineering
Phase 2	Model development and pilot testing
Phase 3	Full deployment with the learning loop
Phase 4	Continuous monitoring and compliance audits

Conclusion

The AI-powered collections system links **data → risk scoring → decisioning → intervention → feedback** into one closed loop. Automation handles scale and speed, humans oversee sensitive and high-impact decisions, and fairness, explainability, and compliance are built in at every stage. Rolled out in four phases, the system targets a **15–25% reduction in 30-day+ delinquency** and a **30% gain in outreach efficiency**, while improving the customer experience.

At a glance

- 4-stage pipeline: ingestion → risk model → decision engine → intervention & feedback
- Retraining every 30–60 days
- AI automates scoring, prioritisation, and standard outreach; humans handle edge cases, approvals, audits, and appeals
- Governance: fairness monitoring, SHAP explainability, human review, regulatory and privacy compliance
- Rollout: audit → pilot → full deployment → continuous monitoring